KL DIVERGENCE

$$\mathrm{KL}(p\|q) = \int p(x)\log\frac{p(x)}{q(x)}\,dx = \mathbb{E}_{x\sim p}\left[\log\frac{p(x)}{q(x)}\right]$$

$\mathrm{KL}\ge 0$; $\mathrm{KL}=0 \Leftrightarrow p=q$; asymmetric.

Univariate Gaussian KL (closed form):

$$\mathrm{KL}\big[\mathcal{N}(\mu_1,\sigma_1)\|\mathcal{N}(\mu_2,\sigma_2)\big] = \log\frac{\sigma_2}{\sigma_1} + \frac{\sigma_1^2+(\mu_1-\mu_2)^2}{2\sigma_2^2} - \frac{1}{2}$$

VAE KL (diag. q vs. $\mathcal{N}(\mathbf{0},\mathbf{I})$):

$$\mathrm{KL}\big[\mathcal{N}(\boldsymbol{\mu},\operatorname{diag}\boldsymbol{\sigma}^2)\|\mathcal{N}(\mathbf{0},\mathbf{I})\big] = \tfrac{1}{2}\textstyle\sum_j\big(\sigma_j^2+\mu_j^2-1-\log\sigma_j^2\big)$$

Multivariate Gaussian KL:

$$\mathrm{KL}(\mathcal{N}_0\|\mathcal{N}_1) = \tfrac{1}{2}\Big[\log\tfrac{|\Sigma_1|}{|\Sigma_0|} + (\boldsymbol{\mu}_0-\boldsymbol{\mu}_1)^\top\Sigma_1^{-1}(\boldsymbol{\mu}_0-\boldsymbol{\mu}_1) + \operatorname{tr}(\Sigma_1^{-1}\Sigma_0) - D\Big]$$

MLE = KL minimisation:

$$\hat\theta = \arg\max_\theta\sum_i\log p_\theta(\mathbf{x}_i) \;\Leftrightarrow\; \arg\min_\theta \mathrm{KL}(p_{\text{data}}\|p_\theta)$$

DLVM / VAE

Generative model: $\mathbf{z}\sim p(\mathbf{z})$, $\mathbf{x}\sim p_\theta(\mathbf{x}|\mathbf{z})$; intractable marginal $p(\mathbf{x})=\int p_\theta(\mathbf{x}|\mathbf{z})p(\mathbf{z})d\mathbf{z}$

ELBO (introduce $q_\phi(\mathbf{z}|\mathbf{x})>0$, apply Jensen $\log\mathbb{E}\ge\mathbb{E}\log$):

$$\log p(\mathbf{x}) = \log\mathbb{E}_{\mathbf{z}\sim q_\phi}\left[\frac{p_\theta(\mathbf{x}|\mathbf{z})p(\mathbf{z})}{q_\phi(\mathbf{z}|\mathbf{x})}\right] \ge \underbrace{\mathbb{E}_{\mathbf{z}\sim q_\phi}[\log p_\theta(\mathbf{x}|\mathbf{z})]}_{\text{reconstruction}} - \underbrace{\mathrm{KL}(q_\phi(\mathbf{z}|\mathbf{x})\|p(\mathbf{z}))}_{\text{regularisation}} =: \mathcal{L}$$

Gap identity:

$$\log p(\mathbf{x}) = \mathcal{L}(\theta,\phi) + \mathrm{KL}\big(q_\phi(\mathbf{z}|\mathbf{x})\|p_\theta(\mathbf{z}|\mathbf{x})\big)$$

Maximising ELBO maximises evidence and minimises KL to true posterior.

Amortised VI (encoder):

$$q_\phi(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\mathbf{z}|\boldsymbol{\mu}_\phi(\mathbf{x}),\operatorname{diag}\boldsymbol{\sigma}_\phi^2(\mathbf{x}))$$

Reparameterisation trick:

$$\mathbf{z} = \boldsymbol{\mu}_\phi(\mathbf{x}) + \boldsymbol{\sigma}_\phi(\mathbf{x})\odot\boldsymbol{\varepsilon},\quad \boldsymbol{\varepsilon}\sim\mathcal{N}(\mathbf{0},\mathbf{I})$$

$$\nabla_\phi\mathcal{L} \approx \nabla_\phi\log p_\theta(\mathbf{x}|\boldsymbol{\mu}_\phi+\boldsymbol{\sigma}_\phi\odot\boldsymbol{\varepsilon}) - \nabla_\phi\mathrm{KL}$$

Low variance. Works for Gaussian/Beta/Gamma; not discrete (use Gumbel-softmax).

VAE Issues

Posterior collapse: $q_\phi(\mathbf{z}|\mathbf{x})\to p(\mathbf{z})$, decoder ignores $\mathbf{z}$. Fix: KL warm-up (β-VAE: weight KL by $\beta\in(0,1)$).

Hole problem: aggregated posterior $q_\phi(\mathbf{z}) = \frac{1}{N}\sum_n q_\phi(\mathbf{z}|\mathbf{x}_n)\ne p(\mathbf{z}) \;\Rightarrow\;$ holes give bad samples.

Improving the Prior

Aggregated posterior (best prior, minimises CE): $q_\phi(\mathbf{z}) = \frac{1}{N}\sum_n q_\phi(\mathbf{z}|\mathbf{x}_n)$

MoG prior: $p_\lambda(\mathbf{z}) = \sum_{k=1}^K w_k\,\mathcal{N}(\mathbf{z}|\boldsymbol{\mu}_k,\operatorname{diag}\boldsymbol{\sigma}_k^2)$

VampPrior: $p_\lambda(\mathbf{z}) = \frac{1}{K}\sum_{k=1}^K q_\phi(\mathbf{z}|\mathbf{u}_k)$, pseudo-inputs $\mathbf{u}_k$ learnable.

Hierarchical VAE (2-level)

Bottom-up: $Q(\mathbf{z}_1,\mathbf{z}_2|\mathbf{x}) = q(\mathbf{z}_2|\mathbf{z}_1)\,q(\mathbf{z}_1|\mathbf{x})$

$$\mathcal{L} = \mathbb{E}[\log p(\mathbf{x}|\mathbf{z}_1)] - \mathbb{E}\Big[\log\tfrac{q(\mathbf{z}_1|\mathbf{x})}{p(\mathbf{z}_1|\mathbf{z}_2)}\Big] - \mathbb{E}[\mathrm{KL}(q(\mathbf{z}_2|\mathbf{z}_1)\|p(\mathbf{z}_2))]$$

NORMALIZING FLOWS

Model: $\mathbf{u}\sim p_\phi(\mathbf{u})$, $\mathbf{x}=T(\mathbf{u})$; T diffeomorphism (bijective, smooth).

Change of variables:

$$p_{\mathbf{x}}(\mathbf{x}) = p_{\mathbf{u}}(T^{-1}(\mathbf{x}))\,|\det J_{T^{-1}}(\mathbf{x})|$$

Inverse function theorem: $J_{T^{-1}} = J_T^{-1}$.

Log-likelihood (MLE):

$$\ell = \sum_i\big[\log p_\phi(T^{-1}(\mathbf{x}_i)) + \log|\det J_{T^{-1}}(\mathbf{x}_i)|\big]$$

Only need T^{-1}, $J_{T^{-1}}$ for MLE; need T for sampling.

Composition $T = T_K\circ\cdots\circ T_1$:

$$\log|\det J_T| = \sum_{k=1}^K\log|\det J_{T_k}(\mathbf{z}_{k-1})|$$

Affine Coupling Layer (RealNVP)

Partition $\mathbf{z}=(\mathbf{z}_{1:d},\mathbf{z}_{d+1:D})$; $s,t:\mathbb{R}^d\to\mathbb{R}^{D-d}$ arbitrary NNs.

Fwd: $\mathbf{z}'_{1:d}=\mathbf{z}_{1:d}$, $\mathbf{z}'_{d+1:D} = e^{s(\mathbf{z}_{1:d})}\odot\mathbf{z}_{d+1:D}+t(\mathbf{z}_{1:d})$

Inv: $\mathbf{z}_{1:d}=\mathbf{z}'_{1:d}$, $\mathbf{z}_{d+1:D} = (\mathbf{z}'_{d+1:D}-t)\odot e^{-s(\mathbf{z}_{1:d})}$

Jacobian (triangular $\Rightarrow$ easy det):

$$J_T = \begin{bmatrix}\mathbf{I} & \mathbf{0}\\ * & \operatorname{diag}(e^{s(\mathbf{z}_{1:d})})\end{bmatrix},\quad \det J_T = \exp\Big(\textstyle\sum_i s_i\Big)$$

Permutation layer: $|\det J| = 1$; needed so all dims transform.

Flow as VAE Prior

$$\mathcal{L} = \mathbb{E}_{\mathbf{z}\sim q_\phi}\big[\log p_\theta(\mathbf{x}|\mathbf{z}) + \log p_\phi(T^{-1}(\mathbf{z}))\big] - \mathbb{E}\big[\log|\det J_T(T^{-1}(\mathbf{z}))|\big] - \mathbb{E}[\log q_\phi]$$

Prefer flow prior over flow posterior (more expressive model, same cost).

DDPM: FORWARD PROCESS

One step: $q(\mathbf{z}_t|\mathbf{z}_{t-1}) = \mathcal{N}(\mathbf{z}_t|\sqrt{1-\beta_t}\,\mathbf{z}_{t-1},\ \beta_t\mathbf{I})$
Marginal (closed form, skips intermediate steps):

$$q(\mathbf{z}_t|\mathbf{x}) = \mathcal{N}\big(\mathbf{z}_t \mid \sqrt{\bar\alpha_t}\,\mathbf{x},\ (1-\bar\alpha_t)\mathbf{I}\big)$$

where $\bar\alpha_t = \prod_{s=1}^{t}(1-\beta_s)$; $\alpha_t = 1-\beta_t$.
Reparametrisation: $\mathbf{z}_t = \sqrt{\bar\alpha_t}\,\mathbf{x} + \sqrt{1-\bar\alpha_t}\,\boldsymbol{\varepsilon}$, $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0},\mathbf{I})$

Proof $q(\mathbf{z}_T|\mathbf{x}) \to \mathcal{N}(\mathbf{0},\mathbf{I})$**:** $\bar\alpha_T < (1-\beta_1)^T \xrightarrow{T\to\infty} 0 \Rightarrow \sqrt{\bar\alpha_T} \to 0$, $1-\bar\alpha_T \to 1$.

DDPM: REVERSE & ELBO

Reverse process (learned): $p_\theta(\mathbf{z}_{t-1}|\mathbf{z}_t) = \mathcal{N}(\mathbf{z}_{t-1}|\boldsymbol{\mu}_\theta(\mathbf{z}_t,t),\ \sigma_t^2\mathbf{I})$
Forward posterior (tractable! Gaussian via Bayes):

$$q(\mathbf{z}_{t-1}|\mathbf{z}_t,\mathbf{x}) = \mathcal{N}(\mathbf{z}_{t-1}|\tilde{\boldsymbol{\mu}}_t,\ \tilde\beta_t\mathbf{I})$$

$$\tilde{\boldsymbol{\mu}}_t = \frac{\sqrt{\bar\alpha_{t-1}}\,\beta_t}{1-\bar\alpha_t}\mathbf{x} + \frac{\sqrt{\alpha_t}(1-\bar\alpha_{t-1})}{1-\bar\alpha_t}\mathbf{z}_t$$

$$\tilde\beta_t = \frac{1-\bar\alpha_{t-1}}{1-\bar\alpha_t}\beta_t$$

ELBO (hierarchical VAE — L_T ignorable, L_0 reconstruction):

$$\log p(\mathbf{x}) \geq -\underbrace{\mathrm{KL}(q(\mathbf{z}_T|\mathbf{x})\|p(\mathbf{z}_T))}_{L_T\approx 0} - \sum_{t=2}^{T}\mathbb{E}_{q(\mathbf{z}_t|\mathbf{x})}\underbrace{[\mathrm{KL}(q(\mathbf{z}_{t-1}|\mathbf{z}_t,\mathbf{x})\|p_\theta)]}_{L_{t-1}} + \underbrace{\mathbb{E}[\log p(\mathbf{x}|\mathbf{z}_1)]}_{L_0}$$

L_{t-1} **closed form** (Gaussian KL, C indep. of θ):

$$L_{t-1} = \frac{1}{2\sigma_t^2}\|\tilde{\boldsymbol{\mu}}_t(\mathbf{z}_t,\mathbf{x}) - \boldsymbol{\mu}_\theta(\mathbf{z}_t,t)\|^2 + C$$

Noise prediction — substituting reparametrisation into $\tilde{\boldsymbol{\mu}}_t$:

$$\tilde{\boldsymbol{\mu}}_t = \frac{1}{\sqrt{\alpha_t}}\left(\mathbf{z}_t - \frac{\beta_t}{\sqrt{1-\bar\alpha_t}}\boldsymbol{\varepsilon}\right) \Rightarrow \text{train } \boldsymbol{\varepsilon}_\theta \approx \boldsymbol{\varepsilon}$$

Simplified loss:

$$L_{t-1} \approx \|\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}_\theta(\sqrt{\bar\alpha_t}\mathbf{x} + \sqrt{1-\bar\alpha_t}\boldsymbol{\varepsilon},\ t)\|^2$$

Training & Sampling

Train: $t \sim \mathcal{U}$; $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0},\mathbf{I})$; $\mathbf{z}_t = \sqrt{\bar\alpha_t}\mathbf{x} + \sqrt{1-\bar\alpha_t}\boldsymbol{\varepsilon}$; gradient step on $\|\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}_\theta(\mathbf{z}_t,t)\|^2$.
Sample: $\mathbf{z}_T \sim \mathcal{N}(\mathbf{0},\mathbf{I})$; for $t = T,\dots,1$; $\boldsymbol{\eta} \sim \mathcal{N}(\mathbf{0},\mathbf{I})$:

$$\mathbf{z}_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(\mathbf{z}_t - \frac{\beta_t}{\sqrt{1-\bar\alpha_t}}\boldsymbol{\varepsilon}_\theta(\mathbf{z}_t,t)\right) + \sqrt{\beta_t}\,\boldsymbol{\eta}$$

SDE-BASED DIFFUSION (VP-SDE)

Forward SDE: $d\mathbf{x} = -\frac{1}{2}\beta(t)\mathbf{x}\,dt + \sqrt{\beta(t)}\,d\mathbf{w}$
Reverse SDE (score $\mathbf{s} = \nabla_{\mathbf{x}_t}\log p(\mathbf{x}_t)$):

$$d\mathbf{x} = \big[f(\mathbf{x},t) - g(t)^2\mathbf{s}\big]dt + g(t)\,d\bar{\mathbf{w}}$$

Probability flow ODE (same marginals, deterministic):

$$\tfrac{d\mathbf{x}}{dt} = f(\mathbf{x},t) - \tfrac{1}{2}g(t)^2\nabla_{\mathbf{x}_t}\log p(\mathbf{x}_t)$$

Denoising Score Matching (train $\mathbf{s}_\theta \approx \nabla\log p$):

$$L_{\mathrm{DSM}} = \mathbb{E}_{t,\mathbf{x}_0,\mathbf{x}_t}\big[\lambda(t)\|\mathbf{s}_\theta(\mathbf{x}_t,t) - \nabla_{\mathbf{x}_t}\log p(\mathbf{x}_t|\mathbf{x}_0)\|^2\big]$$

Linear SDE $\Rightarrow$ $p(\mathbf{x}_t|\mathbf{x}_0)$ Gaussian $\Rightarrow$ tractable.
Key: $\nabla_{\mathbf{x}_t}\log\mathcal{N}(\mathbf{x}_t|\boldsymbol{\mu},\sigma^2\mathbf{I}) = (\boldsymbol{\mu}-\mathbf{x}_t)/\sigma^2 = -\boldsymbol{\varepsilon}/\sigma$

FID

Embed via Inception-v3; fit $\mathcal{N}(\boldsymbol{\mu}_r,\Sigma_r)$, $\mathcal{N}(\boldsymbol{\mu}_g,\Sigma_g)$:

$$\mathrm{FID} = \|\boldsymbol{\mu}_r - \boldsymbol{\mu}_g\|^2 + \mathrm{Tr}\Big(\Sigma_r + \Sigma_g - 2(\Sigma_r\Sigma_g)^{1/2}\Big)$$

Lower = better. Fréchet (Wasserstein-2) distance between Gaussians.

Model Comparison

Model	Likelihood	Inv.	Bottleneck	Speed
VAE	Approx	No	Yes	Fast
Flow	Exact	Yes	No	Fast
DDPM	Approx	No	No	Slow
SBGM	$\sim$Ex.	No/Yes	No	Slow

MODULE 2: RIEMANNIAN GEOMETRY

Identifiability: p_θ identifiable $\Leftrightarrow$ $\theta \mapsto p_\theta$ injective.
VAEs/NNs fail: reparametrise $\mathbf{z}$ without changing $p_\theta(\mathbf{x})$.
Manifold: $\mathcal{M} = f(\mathcal{Z})$, $f: \mathcal{Z}\subseteq\mathbb{R}^d \to \mathbb{R}^D$ smooth, $D > d$.
Immersed: J_f full rank $\forall\mathbf{x}$. (Embedded $\Rightarrow$ immersed, not vice versa.)
MLP with full-rank weights + smooth activations $\Rightarrow$ immersed.
Pullback Riemannian metric:

$$M(\mathbf{x}) = J_f(\mathbf{x})^\top J_f(\mathbf{x}) \in \mathbb{R}^{d\times d}$$

Inner product: $\langle\mathbf{u},\mathbf{v}\rangle_{\mathbf{x}} = \mathbf{u}^\top M(\mathbf{x})\mathbf{v}$.
Invariant to reparametrisations $\Rightarrow$ identifiable distances.

Curve length:

$$L(c) = \int_0^1\sqrt{\dot c^\top M(c)\,\dot c}\;dt$$

Curve energy (Cauchy-Schwarz $\Rightarrow$ $L(c)^2 \leq E(c)$, equality iff const. speed):

$$E(c) = \int_0^1 \dot c^\top M(c)\,\dot c\;dt$$

Constant-speed geodesic: energy minimiser with $\|\dot c\|_M = \text{const.}$

$$L(c)^2 \leq E(c)\ \text{(C-S), equality iff const. speed}$$

$$\text{Geodesic ODE:}\quad \ddot c_i + \sum_{j,k}\Gamma^i_{jk}\dot c_j\dot c_k = 0$$

$\Gamma^i_{jk} = \frac{1}{2}M^{im}(\partial_j M_{mk} + \partial_k M_{mj} - \partial_m M_{jk})$ (Christoffel symbols).
Boundary value (start+end) $\to$ may not be unique.
Initial value (point+velocity) $\to$ unique (Picard-Lindelöf).
Log map (Riemannian subtraction):
$\mathrm{Log}_p(q)$ = initial velocity of geodesic $p \to q$ (tangent vector at p); $\|\mathrm{Log}_p(q)\|_{M(p)} = \mathrm{dist}(p,q)$.
Exp map (Riemannian addition):
$\mathrm{Exp}_p(\mathbf{v})$ = endpoint of geodesic from p with init. velocity $\mathbf{v}$.
Noisy Manifold (Stochastic Decoder)
$p_\theta(\mathbf{x}|\mathbf{z}) = \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_\theta(\mathbf{z}), \mathrm{diag}\,\boldsymbol{\sigma}^2_\theta(\mathbf{z}))$; rewrite as random projection $\mathbf{x} = \boldsymbol{\mu}_\theta(\mathbf{z}) + \Lambda_\theta^{1/2}\boldsymbol{\eta}$.

$$M(\mathbf{z}) = J_{\boldsymbol{\mu}}^\top J_{\boldsymbol{\mu}} + \underbrace{\mathbb{E}[J_{\Lambda^{1/2}}^\top J_{\Lambda^{1/2}}]}_{\text{uncertainty term}}$$

Uncertainty creates "walls" $\Rightarrow$ geodesics follow data.
Ensemble: K decoders; uncertainty = variance across models.
Information Geometry (Fisher-Rao)
KL is locally quadratic in $d\theta$ (2nd-order Taylor):

$$\mathrm{KL}(p(\cdot;\theta)\|p(\cdot;\theta+d\theta)) \approx \tfrac{1}{2}d\theta^\top F(\theta)\,d\theta$$

$$F(\theta)_{ij} = \mathbb{E}\big[\partial_i\log p\cdot\partial_j\log p\big]\ \text{(Fisher info. matrix)}$$

Fisher-Rao metric = Riemannian metric on distribution space; reparametrisation invariant.
Parameter Manifolds
$M(\theta) = \mathbb{E}_{\mathbf{x}}\big[J_f(\mathbf{x};\theta)^\top J_f(\mathbf{x};\theta)\big]$
Kernel manifold: zero-length curves (same function, diff. params).
Image manifold: positive-length curves (diff. functions).
UQ: project Gaussian param samples onto kernel $\Rightarrow$ interpolates data.

GRAPH DEFINITIONS

$G = (V, E)$; adjacency $A \in \{0,1\}^{N\times N}$; degree $D = \mathrm{diag}(d_1, \dots, d_N)$.

Laplacians:

$$L = D - A, \quad L_{\mathrm{sym}} = D^{-\frac{1}{2}} L D^{-\frac{1}{2}} = I - D^{-\frac{1}{2}} A D^{-\frac{1}{2}}$$

$$L_{\mathrm{rw}} = D^{-1}L = I - D^{-1}A$$

All non-negative eigenvalues.

GRAPH STATISTICS

Eigenvector Centrality

$$\lambda \mathbf{e} = A\mathbf{e} \qquad e_u = \frac{1}{\lambda} \sum_{v \in \mathcal{N}(u)} e_v$$

Perron-Frobenius: largest eigenvalue is unique; eigenvector non-negative.

Clustering Coefficient

$$c_u = \frac{|\{(v_1, v_2) \in E : v_1, v_2 \in \mathcal{N}(u)\}|}{\binom{d_u}{2}}$$

$\mathbf{c} = \big(D(D-I)\big)^{-1}\mathrm{diag}(A^3)$; $c_u = [A^3]_{uu}/(d_u(d_u-1))$.

Weisfeiler-Lehman (WL) Test

1. Init: $\ell_v^{(0)} = d_v$ (degree)
2. Update: $\ell_v^{(i)} = \mathrm{hash}(\{\{\ell_u^{(i-1)} : u \in \mathcal{N}(v)\}\})$
3. Summarise: $\mathrm{hash}(\{\{\ell_v^{(i)} : v \in V\}\})$
4. Different summaries $\Rightarrow$ not isomorphic (can fail for some pairs).

MESSAGE PASSING

$$\mathbf{m}_u = \mathrm{agg}^{(k)}(\{\mathbf{h}_v^{(k)} : v \in \mathcal{N}(u)\})$$

$$\mathbf{h}_u^{(k+1)} = \mathrm{update}^{(k)}(\mathbf{h}_u^{(k)}, \mathbf{m}_u)$$

Basic GNN:

$$\mathbf{h}_u^{(k)} = \sigma\Big(W_s \mathbf{h}_u^{(k-1)} + W_n \sum_{v \in \mathcal{N}(u)} \mathbf{h}_v^{(k-1)} + \mathbf{b}\Big)$$

Degree-normalised: $\mathbf{m} = \sum_v \mathbf{h}_v / |\mathcal{N}(u)|$ or $\sum_v \mathbf{h}_v / \sqrt{|\mathcal{N}(u)||\mathcal{N}(v)|}$

Set pooling (universal): $\mathbf{m} = \mathrm{MLP}_\theta(\sum_v \mathrm{MLP}_\phi(\mathbf{h}_v))$

Attention: $\alpha_{u,v} = \mathrm{softmax}(\mathbf{h}_u^\top W \mathbf{h}_v)$, $\mathbf{m} = \sum_v \alpha_{u,v} \mathbf{h}_v$

Scaled dot-product: $\alpha_{u,v} = \mathrm{softmax}(\mathbf{k}_u^\top \mathbf{q}_v / \sqrt{d})$, $\mathbf{m} = \sum_v \alpha_{u,v} \mathbf{v}_v$

GRU update (reduce over-smoothing):

$$\mathbf{r} = \sigma(W_{mr}\mathbf{m} + W_{hr}\mathbf{h} + \mathbf{b}_r)$$

$$\mathbf{z} = \sigma(W_{mz}\mathbf{m} + W_{hz}\mathbf{h} + \mathbf{b}_z)$$

$$\tilde{\mathbf{h}} = \tanh(W_{mh}\mathbf{m} + W_{hh}(\mathbf{r} \odot \mathbf{h}) + \mathbf{b}_h)$$

$$\mathbf{h}^{\mathrm{new}} = \mathbf{z} \odot \tilde{\mathbf{h}} + (1-\mathbf{z}) \odot \mathbf{h}$$

Residual: $\mathbf{h}^{\mathrm{new}} = \mathbf{h} + \mathrm{update}(\mathbf{h}, \mathbf{m})$

NODE EMBEDDINGS (SHALLOW)

Encoder (lookup table): $\mathrm{enc}(u) = \mathbf{z}_u = Z^\top \mathbf{s}_u$ ($\mathbf{s}_u$ one-hot). Transductive only.

Decoders: $\mathbf{z}_u^\top \mathbf{z}_v$; $\sigma(\mathbf{z}_u^\top \mathbf{z}_v + b)$; $e^{\mathbf{z}_u^\top \mathbf{z}_v} / \sum_w e^{\mathbf{z}_u^\top \mathbf{z}_w}$; $\|\mathbf{z}_u - \mathbf{z}_v\|^2$

MSE loss:

$$L_{\mathrm{MSE}} = \sum_{(u,v)} (S_{uv} - \mathbf{z}_u^\top \mathbf{z}_v)^2 = \|S - ZZ^\top\|_F^2$$

Binary cross-entropy (note $1 - \sigma(x) = \sigma(-x)$):

$$L_{\mathrm{BCE}} = -\sum_{(u,v)\in E} \log \sigma(\mathbf{z}_u^\top \mathbf{z}_v + b) - \sum_{(u,v)\notin E} \log \sigma(-\mathbf{z}_u^\top \mathbf{z}_v - b)$$

Random walk (neg. sampling):

$$L_{\mathrm{RW}} = \sum_{(u,v)\in W} \Big[-\log \sigma(\mathbf{z}_u^\top \mathbf{z}_v + b) - \gamma \mathbb{E}_w [\log \sigma(-\mathbf{z}_u^\top \mathbf{z}_w - b)] \Big]$$

GRAPH FOURIER & CONVOLUTIONS

Graph convolution (adjacency as shift):

$$\mathbf{y} = \Big(\sum_{k=0}^{K} h[k]\, A^k\Big)\mathbf{x}$$

DFT: $\tilde{x}[k] = \sum_n x[n]\, \omega_N^{kn}$, $\omega_N = e^{-i2\pi/N}$

Matrix form: $\tilde{\mathbf{x}} = U^H \mathbf{x}$, $\mathbf{x} = U\tilde{\mathbf{x}}$, U unitary ($U^H U = I$).

Convolution theorem (circ. conv. $\Leftrightarrow$ freq. mult.):

$$\mathbf{y} = H\mathbf{x} \;\Leftrightarrow\; \tilde{\mathbf{y}} = \tilde{\mathbf{h}} \odot \tilde{\mathbf{x}}$$

Proof:

$$\tilde{y}[k] = \sum_n \sum_\ell x[\ell]\, h[(n-\ell) \bmod N]\, \omega_N^{kn}$$

$$= \sum_\ell x[\ell] \omega_N^{k\ell} \underbrace{\sum_m h[m] \omega_N^{km}}_{\tilde{h}[k]} = \tilde{x}[k] \tilde{h}[k]$$

Graph Fourier: $A = U \Lambda U^H$. Cycle graph eigenvectors = DFT basis.

Graph conv. via Fourier:

$$\mathbf{y} = U\Big(\sum_{k=0}^{K} h[k] \Lambda^k\Big) U^H \mathbf{x}$$

GNNs AND PROBABILISTIC MODELS

Markov Random Field:

$$p(\{x_v\}, \{z_v\}) \propto \prod_v \Phi(x_v, z_v) \prod_{(u,v)\in E} \Psi(z_u, z_v)$$

Mean field VI (factored $q = \prod_v q_v$, min KL — fixed point):

$$\log q_v^{(t+1)} = c_v + \log \Phi(x_v, z_v) + \sum_{u \in \mathcal{N}(v)} \int q_u^{(t)} \log \Psi \, dz_u$$

This IS message passing; kernel embeddings $\Rightarrow$ GNN updates.

GRAPH VAE

Node-level VAE (encoder = GNN):

$$\boldsymbol{\mu}_Z, \log \boldsymbol{\sigma}_Z = \mathrm{GNN}(A, X), \; Z = \boldsymbol{\mu}_Z + \boldsymbol{\varepsilon} \odot \boldsymbol{\sigma}_Z$$

$$P(A_{uv} = 1|Z) = \sigma(\mathbf{z}_u^\top \mathbf{z}_v + b), \quad p(Z) = \mathcal{N}(\mathbf{0}, \mathbf{I})$$

$$\mathcal{L} = \mathbb{E}_q[\textstyle\sum_{u,v} \log p_\theta(A_{uv}|Z)] - \mathrm{KL}[q_\phi(Z|G) \| p(Z)]$$

Graph-level VAE: $\mu_z, \log \sigma_z = \mathrm{GNN\_readout}(A, X)$; $z = \mu_z + \varepsilon \sigma_z$; $\tilde{A} = \sigma(\mathrm{MLP}(z))$.

GAN: $\min_\theta \max_\phi \mathbb{E}_x[\log(1 - d_\phi(x))] + \mathbb{E}_z[\log d_\phi(g_\theta(z))]$

Generator: $\tilde{A} = \sigma(\mathrm{MLP}(z))$; Discriminator: permutation-invariant GNN.

SYMMETRY IN GEOMETRIC GNNs

Invariant: $f(T(\mathbf{h})) = f(\mathbf{h})$ **Equivariant:** $f(T(\mathbf{h})) = T'(f(\mathbf{h}))$

Inv. ops: $\|\mathbf{v}\|$, $\mathbf{v}_1 \cdot \mathbf{v}_2$, $\|\mathbf{v}_1 \times \mathbf{v}_2\|$ Equiv. ops: $a\mathbf{v}$, $\mathbf{v}_1 + \mathbf{v}_2$, $\mathbf{v}_1 \times \mathbf{v}_2$

EVALUATION OF GRAPH GENERATION

Total variation: $d(s_{\mathrm{gen}}, s_{\mathrm{test}}) = \sup_x |P(s_{\mathrm{gen}} \in x) - P(s_{\mathrm{test}} \in x)|$

Discrete: $d = \frac{1}{2}\|p_{\mathrm{gen}} - p_{\mathrm{test}}\|_1$. Compare: degree, clust. coeff., centrality.

Traditional models:

- **Erdős-Rényi:** $P(A_{uv} = 1) = r$ (edges i.i.d.)
- **Stochastic block model:** nodes $\rightarrow$ blocks; $P(A_{uv} = 1) = r_{ij}$
- **Preferential attachment:** $P(A_{uv} = 1) \propto d_v^{(t)}$ (rich-get-richer)

KEY IDENTITIES

Jensen: concave g: $g(\mathbb{E}[X]) \geq \mathbb{E}[g(X)]$ (e.g. $g = \log$)

Cauchy-Schwarz: $(\int f \cdot g)^2 \leq (\int f^2)(\int g^2)$; equality iff $f \propto g \Rightarrow L(c)^2 \leq E(c)$

PPCA bias: $\hat{\mathbf{b}} = \bar{\mathbf{x}}$ (set $\partial \ell / \partial \mathbf{b} = 0 \Rightarrow C^{-1}(\mathbf{b} - \bar{\mathbf{x}}) = 0$)

Triangular det: $\det A = \prod_i a_{ii}$

Jacobian chain rule: $J_{T \circ S}(x) = J_T(S(x)) \cdot J_S(x)$

Gaussian score: $\nabla_{\mathbf{x}} \log \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \sigma^2 I) = (\boldsymbol{\mu} - \mathbf{x})/\sigma^2 = -\boldsymbol{\varepsilon}/\sigma$

LOTUS: $\mathbb{E}[f(X)] = \int f(x)p(x)dx$ (sum over $t \rightarrow$ expectation $\sim \mathcal{U}$)

Bayes on forward: $q(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{x}) = q(\mathbf{z}_t|\mathbf{z}_{t-1})\, q(\mathbf{z}_{t-1}|\mathbf{x}) / q(\mathbf{z}_t|\mathbf{x})$ (Gaussian, closed form)

Transduction vs induction: Shallow = transductive. GNN = inductive.